COURSE 2 · FORMULA CHEAT SHEET · 1 OF 2

DAX for design engineering data

Table names = the workbook sheets; the cleaned change log is Changes; the date table is Dates; measures live in _Measures.

Dates table (S07), then Mark as date table

```
Dates =
ADDCOLUMNS (
    CALENDAR ( DATE ( 2024, 1, 1 ),
               DATE ( 2026, 12, 31 ) ),
    "Year", YEAR ( [Date] ),
    "MonthNo", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMM" ),
    "YearMonth", FORMAT ( [Date], "YYYY-MM" ),
    "Quarter", "Q" & QUARTER ( [Date] )
)
MonthStart = DATE ( YEAR ( Dates[Date] ),
                    MONTH ( Dates[Date] ), 1 )
```

USERELATIONSHIP: the inactive CloseDate line (S07)

```
Change Count = COUNTROWS ( Changes )
-- active line: Dates[Date] -> Changes[RequestDate]
Changes Closed =
CALCULATE (
    [Change Count],
    USERELATIONSHIP ( Changes[CloseDate], Dates[Date] ),
    NOT ISBLANK ( Changes[CloseDate] )
)
```

DIVIDE: safe ratio, blank instead of an error (S04, S07)

```
Closed % = DIVIDE ( [Changes Closed], [Change Count] )
Open Changes = [Change Count] - [Changes Closed]
```

DATEDIFF: lateness, slip and cycle days (S03, S04, S08)

```
DaysLate =
DATEDIFF ( Drawings[PlannedRelease],
           Drawings[ActualRelease], DAY )
Slip Days =
MAX ( 0, NPD_Stages[Actual Days] - NPD_Stages[Plan Days] )
CycleDays =
IF ( NOT ISBLANK ( Changes[CloseDate] ),
     DATEDIFF ( Changes[RequestDate],
                Changes[CloseDate], DAY ) )
```

SUMX + RELATED: BOM roll-up (B2)

```
Total Target Mass =
SUMX ( BOM,
    BOM[QtyPer] * RELATED ( Parts[TargetMass_g] ) )
```

Rev current on a date (S02, AsOfDate slicer)

```
AsOfDate = CALENDAR ( DATE ( 2024, 10, 1 ),
                      DATE ( 2026, 9, 30 ) )
Current Rev =
VAR AsOf = MAX ( AsOfDate[Date] )
VAR LastRelease =
    CALCULATE ( MAX ( Revisions[ReleaseDate] ),
        Revisions[ReleaseDate] <= AsOf )
RETURN
    IF ( HASONEVALUE ( Parts[PartID] )
             && NOT ISEMPTY ( BOM ),
        CALCULATE ( SELECTEDVALUE ( Revisions[Rev] ),
            Revisions[ReleaseDate] = LastRelease ) )
```

Worst case and RSS stack (S05)

```
WC Tol =
VAR b = SELECTEDVALUE ( SeatRingBuild[Build], "A/B" )
RETURN
    SUMX (
        FILTER ( StackContributors,
            ISBLANK ( StackContributors[Rev] )
                || StackContributors[Rev] = b ),
        StackContributors[Tol_mm] )
RSS Tol: same VAR and FILTER, but
    SQRT ( SUMX ( FILTER ( ... ), [Tol_mm] ^ 2 ) )
Stack Nominal: SUMX ( FILTER ( ... ),
    [Direction] * [Nominal_mm] )
WC Min = [Stack Nominal] - [WC Tol]
RSS Max = [Stack Nominal] + [RSS Tol]
```

Safety factor and Cpk flags (S05)

```
SF = DIVIDE ( FEA[Allowable_MPa], FEA[MaxStress_MPa] )
SF Target = 1.5
Cases Below Target =
COUNTROWS ( FILTER ( FEA, FEA[SF] < [SF Target] ) )
Machines Below 1.33 =
COUNTROWS ( FILTER ( ManufacturingCpk,
    ManufacturingCpk[Cpk] < 1.33 ) )
```

MEDIANX and PERCENTILEX.INC (S08)

```
Median Cycle Days =
MEDIANX (
    FILTER ( Changes, Changes[Status] = "Closed" ),
    Changes[CycleDays] )
P90 Cycle Days =
PERCENTILEX.INC (
    FILTER ( Changes, Changes[Status] = "Closed" ),
    Changes[CycleDays], 0.9 )
```

As-of open backlog (S08)

```
Open Backlog =
VAR d = MAX ( Dates[Date] )
RETURN
IF ( MIN ( Dates[Date] ) > DATE ( 2026, 9, 30 ), BLANK (),
    CALCULATE (
        COUNTROWS ( Changes ),
        REMOVEFILTERS ( Dates ),
        Changes[RequestDate] <= d
            && ( ISBLANK ( Changes[CloseDate] )
                 || Changes[CloseDate] > d ) ) )
```

Bridge table: two counts per part (S07)

```
Changes Affecting Part =
DISTINCTCOUNT ( ChangeAffectedItems[ChangeID] )
Changes by Primary Part =
CALCULATE ( [Change Count],
    USERELATIONSHIP ( Changes[PrimaryPart],
                      Parts[PartID] ) )
```

Field ppm, matured cohorts, Poisson 95 % CI (S09)

```
Matured = Builds[BuildMonth] + 120 <= DATE ( 2026, 9, 30 )
Matured Units =
CALCULATE ( SUM ( Builds[UnitsBuilt] ),
    Builds[Matured] = TRUE () )
Wear Failures 90d =
CALCULATE ( COUNTROWS ( WarrantyClaims ),
    WarrantyClaims[IsSeatRingWear] = TRUE (),
    WarrantyClaims[Within90] = TRUE (),
    WarrantyClaims[Matured] = TRUE () )
Field ppm =
DIVIDE ( [Wear Failures 90d], [Matured Units] ) * 1000000
CI Low ppm =
VAR k = [Wear Failures 90d]  VAR u = [Matured Units]
RETURN IF ( u > 0, IF ( k = 0, 0,
    CHISQ.INV ( 0.025, 2 * k ) / 2 ) / u * 1000000 )
CI High ppm = ... CHISQ.INV ( 0.975, 2 * k + 2 ) / 2
    / u * 1000000
```

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More DAX, Power Query (M) steps, engineering definitions

DAX from Sections 4 and 10; M from Section 6; definitions used in Sections 3 to 5 and 8 to 10.

Pareto cumulative % (S04)

```
Findings Cum % =
VAR cur = [Finding Count]
VAR cats = ALLSELECTED ( ReviewFindings[Category] )
VAR upto =
    SUMX ( FILTER ( cats,
                CALCULATE ( [Finding Count] ) >= cur ),
           CALCULATE ( [Finding Count] ) )
RETURN DIVIDE ( upto, CALCULATE ( [Finding Count], cats ) )
```

RPN before / after (S10)

```
RPN Before = DFMEA[S] * DFMEA[O] * DFMEA[D]
RPN After =
    DFMEA[S_after] * DFMEA[O_after] * DFMEA[D_after]
HighS NoAction =
DFMEA[S] >= 7
    && ( ISBLANK ( DFMEA[Action] ) || DFMEA[Action] = "" )
```

M: folder path parameter ChangeFolder (S06)

```
// Home > Manage Parameters > New: Name ChangeFolder, Type
Text
"<course data folder>\Change_Exports" meta
  [IsParameterQuery = true, Type = "Text",
   IsParameterQueryRequired = true]
```

M: combine every CSV in the folder (S06)

```
// Folder > Combine & Transform Data writes the helper
// queries; then edit the Source step of Changes and of
// Sample File to use the parameter:
Source = Folder.Files ( ChangeFolder )
// result: 21 files -> 184 rows, Source.Name first
```

M: types with an explicit locale (S06)

```
#"Set dates (en-US)" = Table.TransformColumnTypes (
    #"Set text types",
    { {"RequestDate", type date}, {"TriageDate", type date},
      {"ApprovalDate", type date}, {"ImplementDate", type
date},
      {"CloseDate", type date} }, "en-US" ),
#"Set cost (en-US)" = Table.TransformColumnTypes (
    #"Set dates (en-US)",
    { {"CostImpact_USD", type number} }, "en-US" ),
```

M: Trim, UPPERCASE, Replace '-' in PrimaryPart (S06)

```
#"Trim part" = Table.TransformColumns ( #"Set cost (en-US)",
    {{"PrimaryPart", Text.Trim, type text}} ),
#"Upper part" = Table.TransformColumns ( #"Trim part",
    {{"PrimaryPart", Text.Upper, type text}} ),
#"Remove hyphen" = Table.ReplaceValue ( #"Upper part",
    "-", "", Replacer.ReplaceText, {"PrimaryPart"} ),
```

M: drop Source.Name, remove exact duplicates (S06)

```
#"Drop Source.Name" = Table.RemoveColumns (
    #"Remove hyphen", {"Source.Name"} ),
#"Remove exact duplicates" =
    Table.Distinct ( #"Drop Source.Name" )  // 184 in, 178
out
```

Engineering definitions

Worst case (WC) vs RSS stack

Gap = sum of Direction x Nominal. WC: gap +/- the sum of all tolerances (every part at its limit at once). RSS: gap +/- the square root of the sum of squared tolerances. RSS assumes each contributor is centred, independent, normal and +/-tol = +/-3 sigma; with 4 or 5 contributors that is optimistic, so practitioners often widen it with an inflation factor. One-sided geometric terms (roundness, runout, position) only use up clearance but are entered as +/-, so the upper end of the STK-01 range is overstated (conservative against the max limit). STK-01 is a diametral clearance: diameter terms are not halved.

Safety factor (SF)

SF = Allowable / MaxStress. Allowable = yield strength for static load cases, endurance limit for fatigue cases. Course target: 1.5.

Cpk

Cpk = min(USL - mean, mean - LSL) / (3 sigma): how far the process mean sits from the nearer limit, in units of 3 sigma. 1.33 is a common minimum (4 sigma to the nearer limit). Show 3 decimals: 1.329 is below 1.33.

Field ppm with a matured cohort

Failures of one mode within 90 days of the build month, per 1,000,000 product units built (not per seat ring). Count only build months with build month + 120 days on or before the data cut-off (30 Sep 2026), so every unit has had its full window plus claim lag. 95 % CI on k failures: CHISQ.INV(0.025, 2k)/2 to CHISQ.INV(0.975, 2k+2)/2, divided by units. Few failures = wide interval = early signal only.

Validation test pass rate

Pass lots / all validation test lots. It is not production first-pass yield.

RPN and Action Priority

RPN = Severity x Occurrence x Detection (each 1 to 10). A detection action should lower D, not O; a design change usually lowers O, not D. RPN treats S, O and D as equal, so a high-severity risk can rank low. The AIAG-VDA FMEA Handbook (2019) replaces RPN ranking with Action Priority (AP: High, Medium, Low), which looks at S first, then O, then D.

ECR vs ECO

ECR = engineering change request (someone asks for a change). ECO = engineering change order (the approved change is carried out). In the course log the ID stays ECR-nnnn for the whole life of a change.

On-time gate / slip days

On-time gate: ActualEnd on or before PlannedEnd. Slip days: max(0, actual days - planned days) for that stage only.

The 12 NPD stages (sheet NPD_Stages)

1. Requirements & specification
2. Preliminary concepts
3. Preliminary drawings
4. Concept finalization (gate)
5. Final 3D modeling
6. Design verification (FEA, tolerance stacks, DFMEA)
7. Detailing (2D drawings, GD&T)
8. Checking
9. Design review
10. Release
11. Prototype & validation
12. Production handover